Lecture 13 // Oct 6 // Defining Instanton Floer Homology

Last time: we restricted to Y^3 a $\mathbb{Z}$HS^3, and fixed the (trivial) SU(2) bundle E over Y. Let $\mathcal{A}^{\text{irr}} \subset \mathcal{A}$ be the subset of connections with stabilizer $\pm I$.

Define

$$\mathcal{B} = \underbrace{\mathcal{A}}_{\text{SU(2) connections}} / \underbrace{\mathcal{G}}_{\text{gauge group}}, \qquad \mathcal{B}^* = \mathcal{A}^{\text{irr}}/\mathcal{G}.$$

Then the Chern-Simons Functional is a map $\text{CS} : \mathcal{B} \to \mathbb{R}/\mathbb{Z}$.

Definition 1 (IFH, irreducible complex)

$$C(Y) = \mathbb{F}\langle \text{crit points of CS on } \mathcal{B}^* \rangle.$$

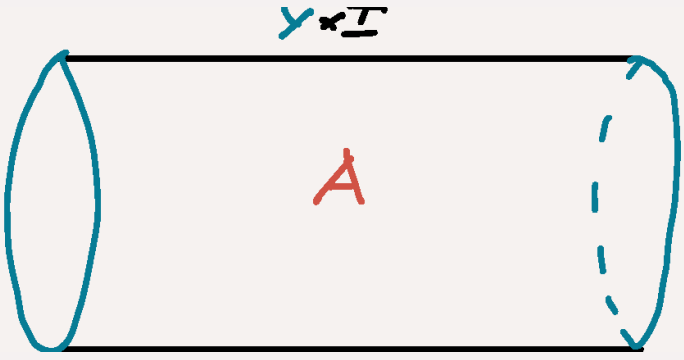
Remark 2
We have to perturb CS to ensure that the critical points are isolated. This perturbation happens in the same way one perturbs a function $f : M \to \mathbb{R}$ to be Morse.

Since $\mathcal{G}$ has $\mathbb{Z}$-many path components, $\pi_0(\mathcal{G}_0) = \mathbb{Z}$, so $\pi_1(\mathcal{B}^*) = \mathbb{Z}$. Now, take $x, y \in \text{Crit(CS)}$ and let γ be a path from x to y in $\mathcal{B}$.

The set of homotopy classes of such paths is $\mathbb{Z}$.

Definition 3
The moduli space of γ-flow lines from x to y is

$$\mathcal{M}_\gamma(x,y) = \left\{ \begin{array}{l} \text{ASD connections on } Y \times I \\ \text{which limit to } x \text{ and } y \text{ on each} \\ \text{end and lie in homotopy class of } \gamma \end{array} \right\} \Big/ \text{ gauge transformations on } Y \times I.$$



Note that there is a free $\mathbb{R}$ action on $\mathcal{M}_\gamma(x,y)$ by translation. If you've seen Morse theory, you know where this is going: we'll first quotient by $\mathbb{R}$, then count flowlines.

Definition 4

$\mathring{\mathcal{M}}_\gamma(x,y) = \mathcal{M}_\gamma(x,y)/\mathbb{R}$,

$$\partial x = \sum_y \sum_{\gamma \in S} \left| \mathring{\mathcal{M}}_\gamma(x,y) \right| y,$$

where S is the set of homotopy classes of paths from x to y for which $\dim(\mathring{\mathcal{M}}_\gamma(x,y)) = 0$.

It turns out that, in contrast to finite-dimensional Morse theory, the $\dim(\mathring{\mathcal{M}}_\gamma(x,y))$ depends on γ: different homotopy classes of path γ from x to y give different dimensional moduli spaces. In particular, the expected dimensions of the ASD moduli spaces differ by 8. Thus $C(Y)$ is a relatively $\mathbb{Z}/8$-graded complex. The relative grading difference between x and y is

$$\mathrm{gr}(x) - \mathrm{gr}(y) = \dim(\mathcal{M}(x,y)) \bmod 8.$$

To get an absolute $\mathbb{Z}/8$ grading on $\mathrm{CS}(Y)$, recall that for Y a $\mathbb{Z}\mathrm{HS}^3$, there is a unique (up to gauge) reducible flat connection on Y. Denote this connection by Θ, and define absolute gradings by counting flows into Θ.

Definition 5

The absolute grading of $x \in \mathrm{Crit}(\mathrm{CS})$ is

$$\mathrm{gr}(x) := \dim(\mathcal{M}(x,\Theta)) \bmod 8.$$

However, something goes wrong when we try to include Θ as a generator of our chain complex. To see this, let's add Θ. Since the boundary map ∂ is only defined for irreducible critical points, we must define new boundary maps for flows into and out of Θ. These are Donaldson's D_1 and D_2 maps.

Definition 6

$D_1 : C(Y) \to \mathbb{F}$ (think of as $\mathbb{F}\langle\Theta\rangle$),

$$D_1(x) = \sum_{\substack{\gamma \text{ from } x \text{ to } \Theta \\ \text{s.t. } \dim(\mathcal{M}(x,\Theta))=1}} |\mathcal{M}(x,\Theta)/\mathbb{R}|\,.$$

Definition 7

$D_2 : \mathbb{F} \to C(Y)$,

$$D_2(1) = \sum_y \left(\sum_{\substack{\gamma \text{ from } \Theta \text{ to } y \\ \text{s.t. } \dim(\mathcal{M}(\Theta,y))=1}} |\mathcal{M}(\Theta,y)/\mathbb{R}|\; y \right).$$

Notice that D_1 is supported on $C_1(Y) = \{x \in C(Y) \text{ s.t. } \mathrm{gr}(x) = 1\} \subset C(Y)$, since the only points which could flow into Θ must have grading 1. However, it turns out that D_2 has image in $C_{(4)}(Y)$. This is strange.

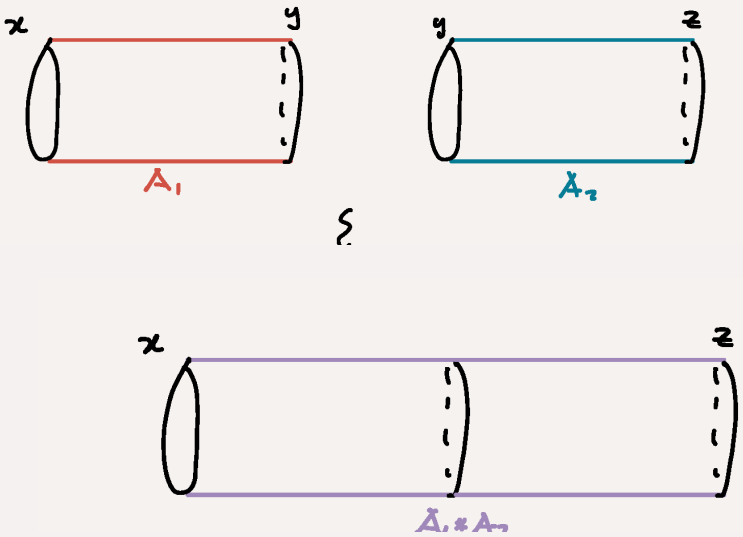
Let's go back to our definition of relative gradings. Why were these well-defined? The implicit claim here was that for $x, y, z \in \mathrm{Crit}(\mathrm{CS})$,

$$\dim(\mathcal{M}(x,y)) + \dim(\mathcal{M}(y,z)) = \dim(\mathcal{M}(x,z)).$$

E.g., we can construct a flow from x to z by gluing flows from $x \to y$ to flows from $y \to z$.

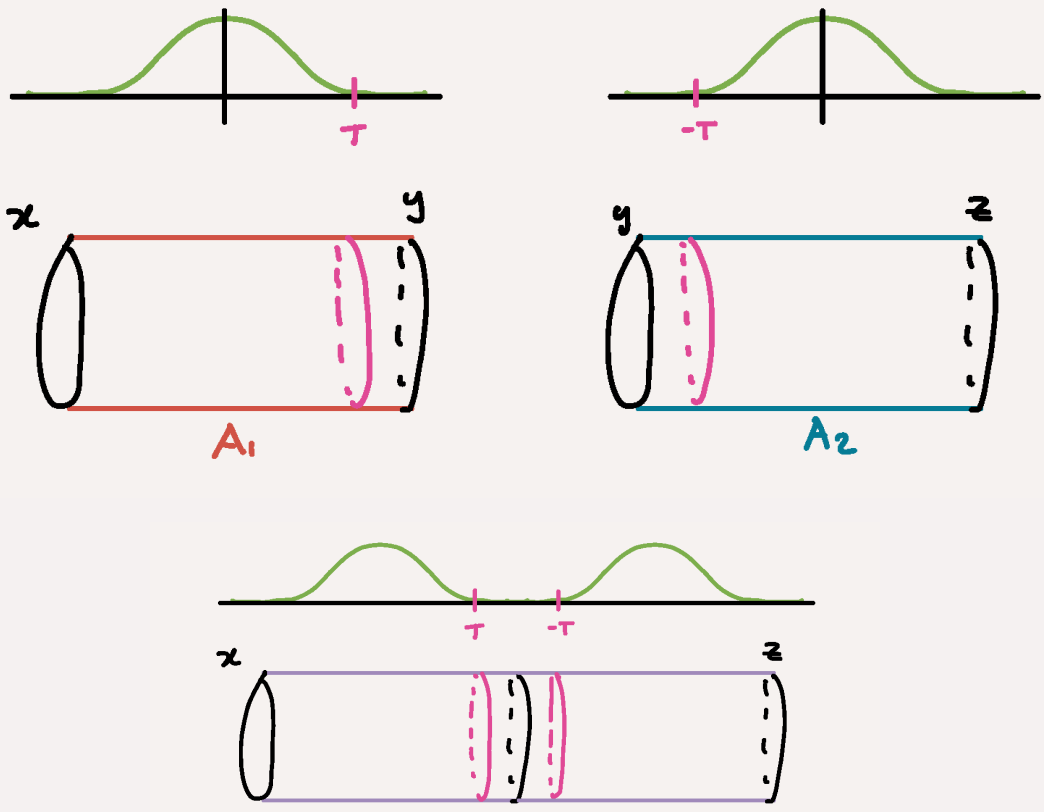


But something goes wrong in the Instanton Floer complex when $y = \Theta$. Say we believe in some gluing procedure. Then, given an ASD connection A_1 on $Y \times \mathbb{R}$ which limits to x as $t \to -\infty$ and y as $t \to \infty$, and similarly an A_2 from y to z, we'd want a connection $A_1 * A_2$ from x to z on $Y \times \mathbb{R}$ built from A_1 and A_2.

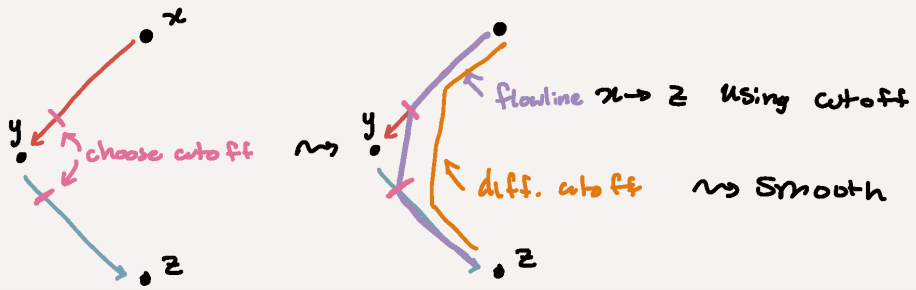


To "glue" A_1 and A_2 together at y to get something smooth, we'll need to add a decay function that ensures that after a certain time T,

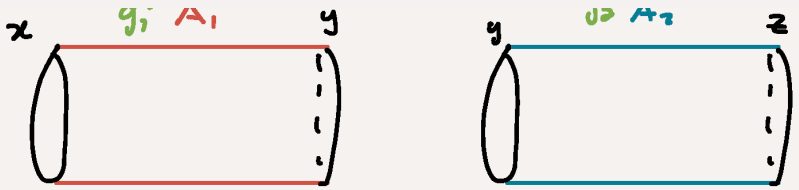
$$f \circ A_1|_{[T,\infty)} = f \circ A_2|_{(-\infty,T]}.$$



This gives us a 1-dimensional gluing parameter T. In the flowline picture, the analog is building a flowline from x to z by choosing a cutoff parameter, following $x \to y$ until then,



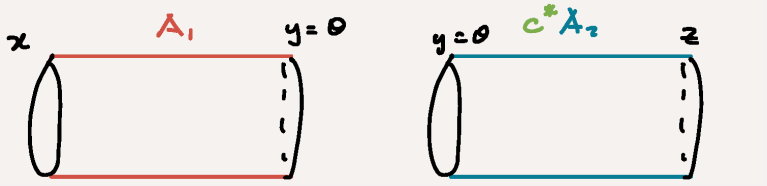
What goes wrong when y has large stabilizer? To glue, we can pick representatives on each end which are literally the same. But we can also compose with gauge transformations to get new gluings.



If $g_1^* A_1|_y = g_2^* A_2|_y$, then the connection on the composite cylinder is gauge-equivalent to the original connection glued via the identity.

But if Stab(y) is big, we get many more connections on $Y \times \mathbb{R}$ than we would expect!

$c =$ constant, non-identity gauge transformation on A_2.



Since y has big stabilizer, $c^*(A_2)|_y = A_2|_y$, so we can still glue to get a connection on the entire composite cylinder. However, $A_1 * c^* A_2$ is not gauge equivalent to $A_1 * A_2$, since any gauge transformation $c^* A_2 \to A_2$ must change A_1 as well. Therefore, we get a $\mathrm{Stab}(y)$'s worth of extra distinct connections on $A_1 * A_2$. Since $y = \Theta$, and $\mathrm{Stab}(\Theta) \cong \mathrm{SU}(2)$,

$$\dim(\mathcal{M}(x,z)) = \dim(\mathcal{M}(x,y)) + \dim(\mathcal{M}(y,z)) + \dim(\mathrm{SU}(2)) = \dim(\mathcal{M}(x,y)) + \dim(\mathcal{M}(y,z)) + 3.$$

This exactly matches what was claimed earlier: for $D_1 : x \to y$, $D_2 : y \to z$, we expect a grading difference of 2 for $x \to z$, but $1 - (-4) = 5 = 2 + 3$.

Another important fact is that in general, $D_2 \circ D_1 \neq 0$. This is in contrast with finite-dimensional Morse theory, where $\partial^2(x) = 0$ for all x. Let's recall why this is true.

$$\partial^2(x) = \partial(\partial(x)) = \partial \begin{pmatrix} \sum_{y \in C_{k-1}} n(x,y)\, x \\ \#\text{flows } x \to y \end{pmatrix} = \sum_{z \in C_{k-2}} \left(\sum_{y \in C_{k-1}} n(x,y)\, n(y,z) \right) y.$$

$\partial^2(x)$ breaks up as a sum of terms cz, where c is the number of broken flowlines from x to z.

$$\partial^2(x) = \sum \cdots + cz + \cdots,$$

The claim is that the coefficient c is always 0. Indeed, the moduli space $\mathcal{M}(x,z)/\mathbb{R}$ is a 2-dimensional manifold/$\mathbb{R}$, so it is an open 1-manifold. It admits a compactification, where the endpoints are broken flowlines. Thus the number of broken flowlines is even. Thus zero mod 2, and with slightly more work, one can show the signed count is also zero.

However, this relies on the fact that $\mathcal{M}(x,z)$ is a 2-dimensional manifold, which comes from the index difference of 2. Therefore this argument fails for Θ in IFH, so we can't add in the reducible and get a chain complex. But $D_1 \circ \partial = 0$ and $\partial \circ D_2 = 0$, so there's some hope of working around this. One such way is by augmenting.

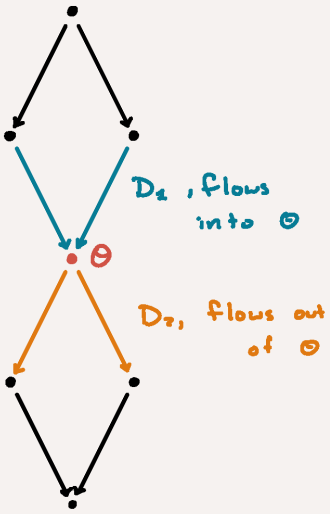
Note that $D_2(\Theta)$ is a cycle in $C(Y)$, and thus defines a homology class $[D_2(\Theta)] \in H_{-4}(C)$. D_1 defines a cohomology element since it vanishes on $\mathrm{im}(\partial)$, so $[D_1] \in H^1(C)$.

Given the irreducible complex C with D_1 and D_2, we can define the reducible complex by either precomposing D_2 or postcomposing D_1 with ∂, i.e., we only allow either flows into or out of Θ.

Definition 8

The "upper bar complex" is $\overline{C} = (C \oplus \mathbb{F}_{(0)},\ \overline{\partial} = \partial + D_1)$.

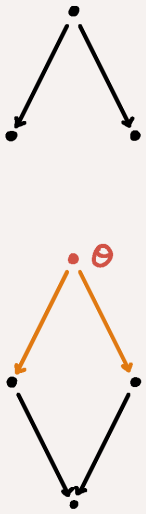
The "lower bar complex" is $\underline{C} = (C \oplus \mathbb{F}_{(-3)},\ \underline{\partial} = \partial + D_2)$.





$\overline{C}$

Cuts out flows out of Θ



$\underline{C}$

Cuts out flows into Θ